

Final project

Themis Oluwatomole Ejegreh \$ patricia givort cruz cabral

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Introduction

1. Network Descriptives

completed

We generate a figure showing network density across waves.

These values are proportions of possible friendships that exist. So for Wave V, only about 14.5% of all possible ties were actually reported. Wave X is slightly denser (about 17.7%), and Wave Y is very similar (about 16.9%). Networks with density < 0.2 are considered sparse, which is very common in social networks because not everyone is friends with everyone else. We generate a table showing basic descriptives of the best-friends network across waves.

Figure 1: Network density across waves

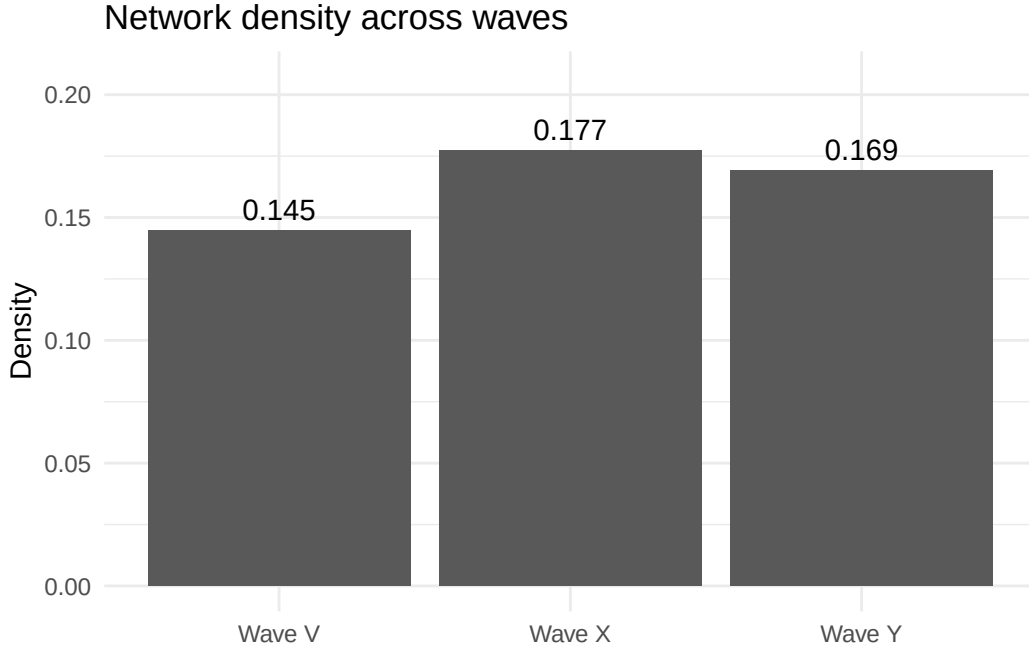


Table 1: Basic descriptives of the best-friends network across waves

Wave	Nodes	Density	DensityChangePct
Wave V	30	0.145	NA
Wave X	28	0.177	22.386
Wave Y	26	0.169	-4.524

Over the three waves of data collection, the selected classroom shows a moderate change in its social structure. The number of students decreased slightly from 30 in Wave V to 28 in Wave X, and then to 26 in Wave Y, indicating some turnover in classroom composition. The density of the “best friends” network increased from 0.145 in Wave V to 0.177 in Wave X (a 22.4% increase), before changing to 0.169 in Wave Y (a -4.5% change relative to Wave X). Overall, the classroom exhibits a relatively stable friendship structure, which is suitable for longitudinal network modelling.

For the selected classroom, the “Best friends” network was constructed from the survey responses, transforming each student’s nominations into an adjacency matrix. Missing codes 98/99 (as documented in the codebook) were treated as missing nominations (i.e., no tie created).

We generate tables showing (i) the Jaccard stability of the node set across waves and (ii) the number of students present in all three waves.

Table 2: Jaccard stability of the node set (classroom composition)

Wave_Comparison	Jaccard_NodeSet
Wave V $\rightarrow$ Wave X	0.933
Wave X $\rightarrow$ Wave Y	0.862

Table 3: Number of students present in all three waves (intersection)

Students_in_all_three_waves
25

The Jaccard indices above quantify how stable the classroom composition is across waves. Because SAOMs require a constant node set, the subsequent SAOM analysis will be run on the students who are present in all three waves (the intersection).

2. Dyadic Hypothesis

For the analysis of dyadic characteristics in the “Best friends” network, I formulated the following hypothesis: students are more likely to nominate as best friends those peers who are of the same sex. This hypothesis is based on the well-established principle of sex homophily in adolescent social networks, where boys tend to form friendships with other boys, and girls with other girls. Theoretical considerations, such as social identity theory, suggest that individuals prefer closer ties with peers who share salient characteristics like sex, which can also facilitate participation in similar classroom activities or interests. Empirical studies in classroom social networks consistently report high levels of sex-based homophily (e.g., Snijders et al., 2010). To operationalize this dyadic hypothesis, a sex similarity matrix was constructed for each wave, where each cell represents whether a pair of students share the same sex (coded as 1) or not (coded as 0). This matrix can later be used in stochastic actor-oriented models (SAOMs) to formally test the influence of sex similarity on the formation and maintenance of friendship ties over time. We generate tables showing 5×5 subsets of the sex similarity matrices for each wave.

Table 4: Sex similarity matrix (Wave V; 5×5 subset)

1	2	3	4	5
1	1	0	1	0
1	1	0	1	0
0	0	1	0	1
1	1	0	1	0
0	0	1	0	1

Table 5: Sex similarity matrix (Wave X; 5×5 subset)

1	2	3	4	5
1	0	1	0	1
0	1	0	1	0
1	0	1	0	1
0	1	0	1	0
1	0	1	0	1

Table 6: Sex similarity matrix (Wave Y; 5×5 subset)

	1	2	3	4	6
1	1	1	0	1	1
2	1	1	0	1	1
3	0	0	1	0	0
4	1	1	0	1	1
6	1	1	0	1	1

To operationalize this dyadic hypothesis, sex similarity matrices were constructed for each wave, where each cell represents whether a pair of students share the same sex (coded as 1) or not (coded as 0). Examples of these matrices for a subset of students are:

Wave V (subset 5×5) 1 2 3 4 5 1 1 1 0 1 0 2 1 1 0 1 0 3 0 0 1 0 1 4 1 1 0 1 0 5 0 0 1 0 1

Wave X (subset 5×5): 1 2 3 4 5 1 1 0 1 0 1 2 0 1 0 1 0 3 1 0 1 0 1 4 0 1 0 1 0 5 1 0 1 0 1

Wave Y (subset 5×5, new student 6 included): 1 2 3 4 6 1 1 1 0 1 1 2 1 1 0 1 1 3 0 0 1 0 0 4 1 1 0 1 1 6 1 1 0 1 1

These matrices can be used in stochastic actor-oriented models (SAOMs) to formally test the influence of sex similarity on the formation and maintenance of friendship ties across waves. The consistent pattern of higher values along clusters of same-sex students supports the theoretical expectation that sex homophily plays a role in the friendship formation in this classroom

3. Triadic Hypothesis

For the analysis of triadic characteristics in the “Best friends” network, I formulated the following hypothesis: friendship ties are likely to exhibit transitivity, meaning that if student A nominates student B as a best friend, and student B nominates student C, then student A is also likely to nominate student C. This hypothesis is grounded in the principle of triadic closure, a common phenomenon in social networks, where “friends of friends tend to become friends” (Granovetter, 1973; Snijders et al., 2010). Theoretical explanations suggest that transitive ties increase trust, reduce social tension, and facilitate cooperation within a group. In classroom networks, transitivity is often observed because students share activities, classes, and social environments, making it more likely that friendship clusters form.

To operationalize this triadic hypothesis, a transitivity measure or a triadic closure matrix can be calculated for each wave. Each cell in such a matrix represents whether a transitive relationship exists between a set of three students (coded as 1 if present, 0 otherwise). These matrices or transitivity statistics can later be included in stochastic actor-oriented models (SAOMs) to formally test whether the presence of a friendship between two students increases the likelihood of forming a friendship with a third connected student, capturing the clustering tendency in the classroom over time

We generate a table showing global transitivity (triadic closure) for each wave.

Table 7: Global transitivity by wave

Wave	Transitivity
Wave V	0.479
Wave X	0.610
Wave Y	0.500

The transitivity of the “Best friends” network, which measures the tendency of friends-of-friends to also be friends (triadic closure), shows a moderate to high level of clustering across the three waves. In Wave V, the transitivity is 0.479, indicating that a substantial share of possible triads are closed. Wave X exhibits a transitivity of 0.61, reflecting the tendency for friendship triads to form. In Wave

Y, the transitivity is 0.5, suggesting that the level of triadic closure remains in a similar range over time. These results support the hypothesis that friendship formation in the classroom is characterized by transitive relationships, consistent with established findings in adolescent social networks where social groups tend to form tightly connected clusters

4. Node Attribute Hypothesis

For the analysis of node-level characteristics in the “Best friends” network, I formulated the following hypothesis: students with higher levels of perceived popularity are more likely to be nominated as best friends by their peers. This hypothesis is based on the idea that individuals with high social status or visibility in the classroom attract more friendship nominations, consistent with the principle of preferential attachment in social networks. Empirical studies in adolescent classrooms indicate that popular students tend to receive more friendship ties and maintain them over time (e.g., Snijders et al., 2010). To operationalize this hypothesis, a popularity score was assigned to each student based on peer-rated measures or the total number of received friendship nominations. This score can then be used as a covariate in stochastic actor-oriented models (SAOMs) to test whether higher popularity increases the likelihood of receiving new nominations and maintaining existing ones over multiple waves

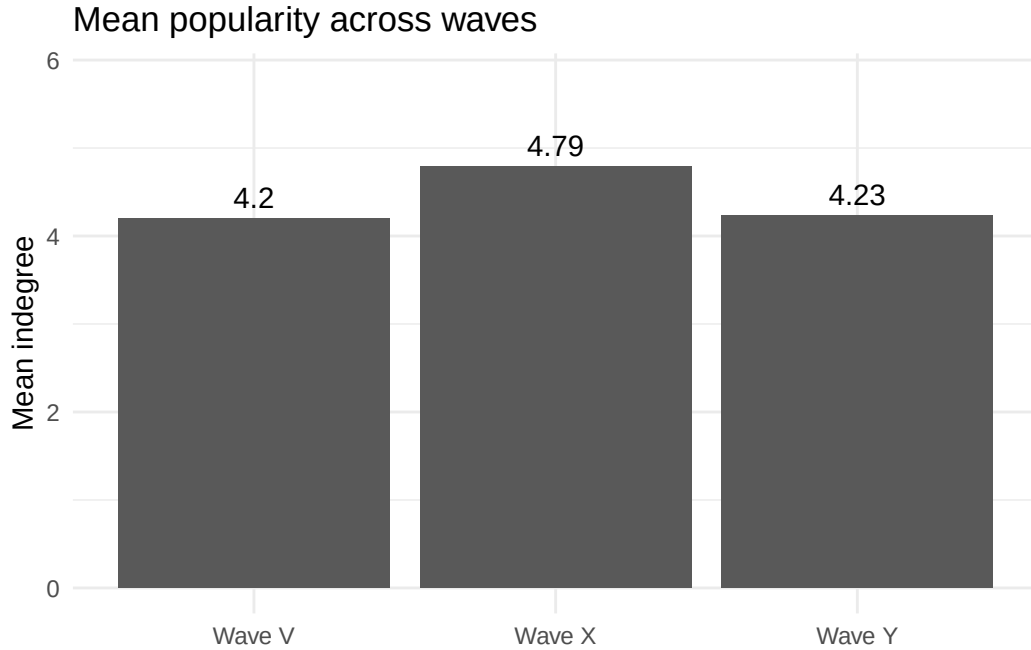
We generate a table and a figure showing popularity (indegree) across waves.

Table 8: Popularity (indegree) summary by wave

Wave	MeanPopularity	MinPopularity	MaxPopularity
Wave V	4.20	0	8
Wave X	4.79	1	8
Wave Y	4.23	1	7

For the node-level analysis, the popularity of each student was calculated as the total number of best-friend nominations they received in each wave. The data were initially transformed into a long format, where each student in each wave was represented as a separate row, resulting in 84 rows in total (30 students in Wave V, 28 in Wave X, and 26 in Wave Y). To summarize the key characteristics of this node-level variable, we computed the mean, minimum, and maximum popularity per wave. In Wave V, students received on average 4.2 nominations, with a range from 0 to 8. In Wave X, the mean popularity was 4.79, with a minimum of 1 and a maximum of 8. In Wave Y, the mean popularity was 4.23, with a minimum of 1 and a maximum of 7. These summaries indicate that while most students receive

Figure 2: Mean popularity (indegree) across waves



several friendship nominations, some students are consistently more popular than others, which may influence the formation and maintenance of best-friend ties over time.

5. Control Variables

Our analysis of the “Best friends” network, several control variables were included to account for factors that might influence friendship formation and maintenance:

Ego-level variables: These describe the characteristics of the student sending the friendship nomination. For example, sex of the ego is included to account for potential homophily effects, where students may prefer to nominate peers of the same sex.

Alter-level variables: These describe characteristics of the student receiving the nomination. For instance, popularity (measured as the number of nominations received) is included to account for the tendency of students to nominate highly popular peers as friends.

Dyadic variables: These variables describe pairwise relationships between students. For example, sex similarity is coded as 1 if both students share the same sex and 0 otherwise, capturing the dyadic homophily effect on friendship nominations.

Structural/network variables: These account for endogenous network processes, such as transitivity, which measures the tendency for friends of friends to become friends, and reciprocity, which captures the tendency for a nomination to be returned by the alter. Including these variables helps model the structural dependencies inherent in social networks.

Together, these control variables allow the stochastic actor-oriented model to isolate the effects of individual, dyadic, and structural factors on the evolution of the friendship network over time.

We generate a table listing the control variables used in the analysis.

Table 9: Control variables to be included in the analysis

Variable	Type	Description
Sex (ego)	Ego-level	Sex of the student sending the nomination, to account for homophily.
Popularity (alter)	Alter-level	Number of nominations received by the student, capturing preference for popular peers.
Sex similarity (dyad)	Dyadic	1 if both students share the same sex, 0 otherwise; captures dyadic homophily.
Transitivity (structural)	Structural	Tendency for friends of friends to become friends; accounts for clustering in the network.
Reciprocity (structural)	Structural	Tendency for a friendship nomination to be reciprocated; captures mutual connections.

6. Longitudinal Plots

We generate figures showing the evolution of the best-friends network across waves (V, X, Y), using **fixed node positions** and **node colors by sex**. Fixed positions make it possible to compare structural change over time without the layout itself moving the nodes around. We focus on three waves (V, X, Y) to follow the recommendation of using at least three waves while keeping a sufficiently stable node set for SAOM estimation. Including Wave W would further reduce the overlap of pupils observed across all waves (intersection), which would decrease sample size and potentially reduce estimation stability.

Figure 3: Best-friends network in Wave V (fixed node positions; node color indicates sex).

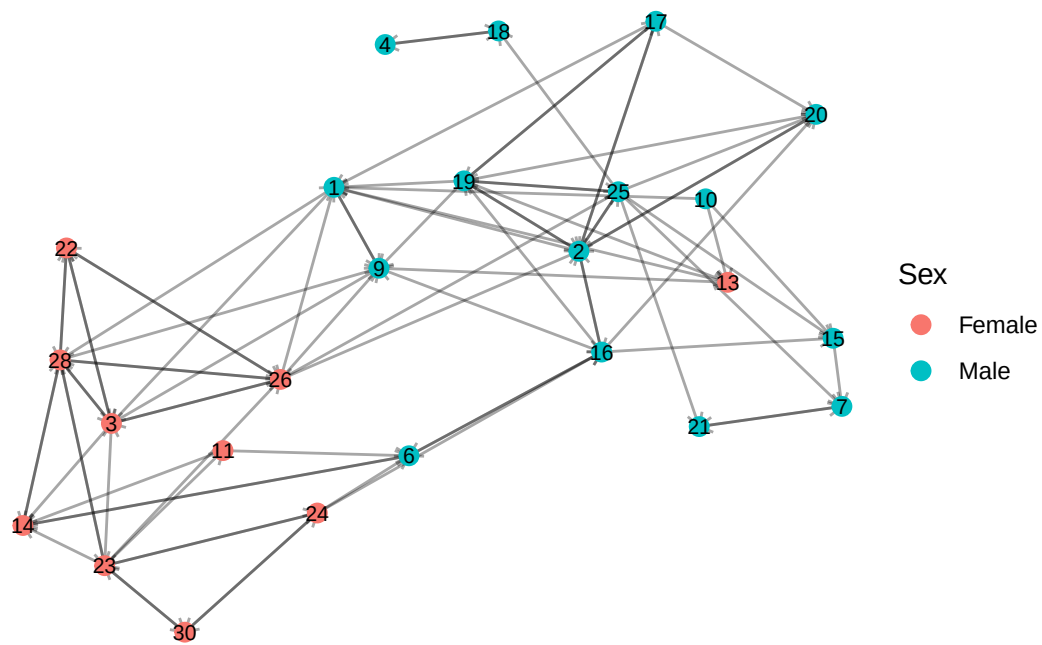


Figure 4: Best-friends network in Wave X (fixed node positions; node color indicates sex).

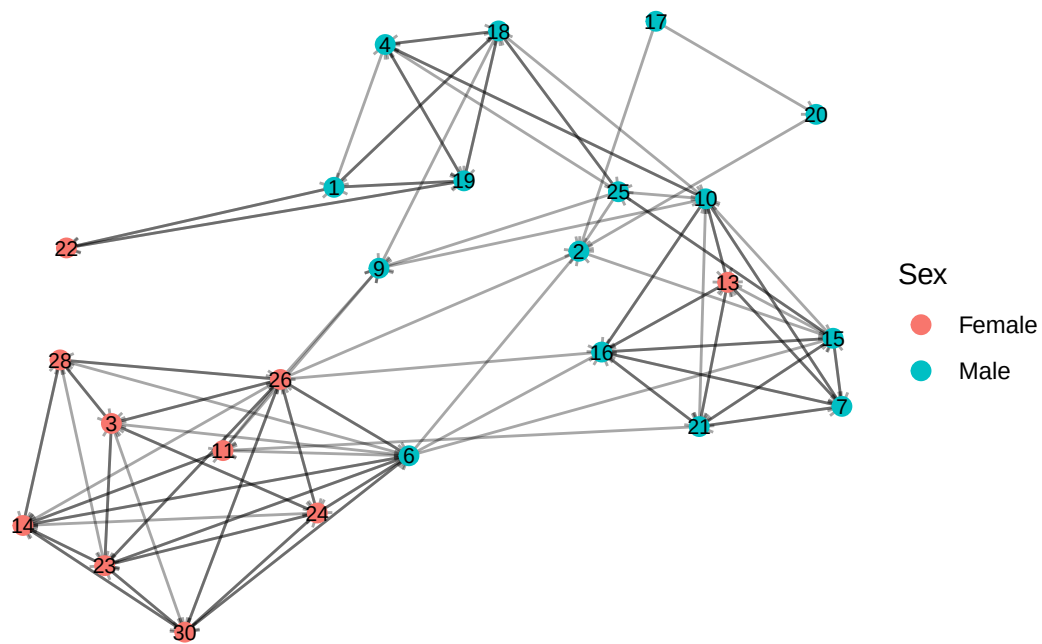
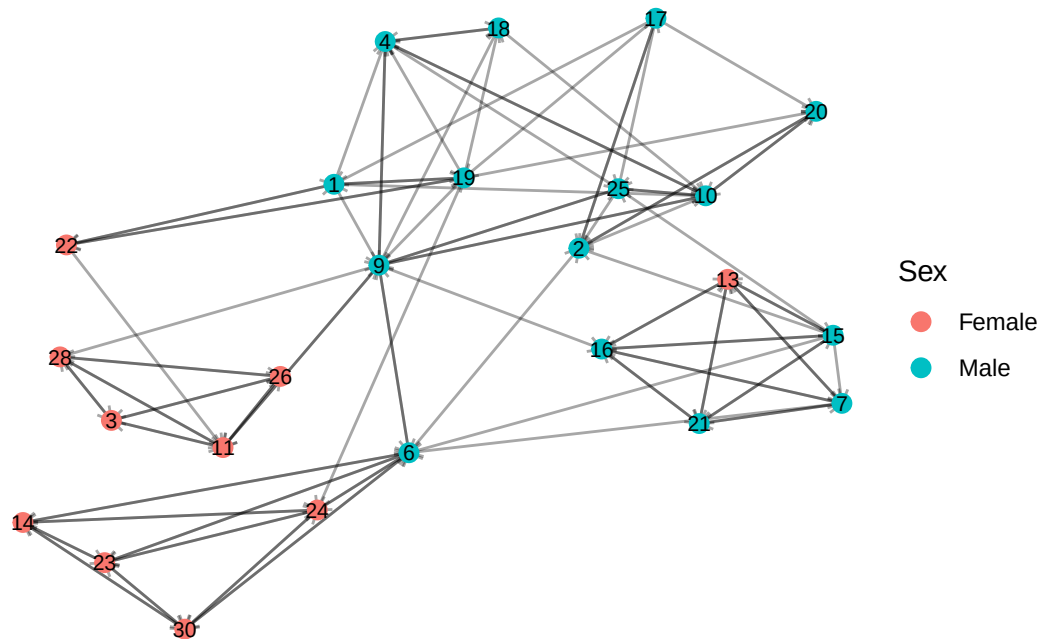


Figure 5: Best-friends network in Wave Y (fixed node positions; node color indicates sex).



7. Model Specification

In this project, we test three hypotheses using a stochastic actor-oriented model (SAOM) in RSiena. We operationalize each hypothesis by selecting the corresponding RSiena effect(s). We generate a table showing how each hypothesis is translated into a SAOM effect. For each hypothesis, we select the corresponding RSiena effect that captures the proposed mechanism of tie formation and maintenance in the best-friends network.

Table 10: Operationalization of hypotheses as RSiena effects.

Hypothesis	RSiena_effect	Ex- pected_di- rection	Interpretation
Dyadic: sex homophily	sameX (interaction1 = sex_cov)	Positive	Ties are more likely between pupils of the same sex.
Triadic: transitivity	transTrip	Positive	Friend-of-a-friend closure: $A \rightarrow B$ and $B \rightarrow C$ increases $A \rightarrow C$.

Hypothesis	RSiena_effect	Ex-pected_di-rection	Interpretation
Node-level: popularity reinforcement	inPop (indegree popularity)	Positive	Those with higher indegree tend to attract additional nominations.

First, sex homophily is modeled with the sameX effect using the sex covariate (interaction1 = “sex_cov”). A positive parameter would indicate that pupils are more likely to nominate same-sex peers as best friends. Second, transitivity is captured by transTrip, where a positive parameter indicates triadic closure (friends-of-friends become friends). Third, popularity reinforcement is modeled with inPop (indegree popularity); a positive parameter indicates a Matthew effect in which pupils who already receive many nominations tend to attract additional nominations over time

8. Model Estimation

We conduct RSiena analyses to test the hypotheses above. Because SAOMs require the same node set across waves, we use the SAOM-aligned matrices (`netV_saom`, `netX_saom`, `netY_saom`) constructed from pupils present in all three waves. We estimate an initial model (Model 1) including our hypothesis-driven effects (sameX, transTrip, inPop) and a standard control for sender heterogeneity (outAct). If convergence is not reached, we increase estimation effort (n3) and continue from the previous estimates using `prevAns`.

We generate a table showing the estimated parameters of the final converged model.

Table 11: Convergence diagnostics for the final estimation run.

Metric	Value
tconv.max	0.122
max(tconv)	0.032

The Table 12 shows the parameter estimates for the final model (Estimate, standard error, and t-ratio).

Table 12: RSiena parameter estimates for the final model.

Effect	Estimate	SE	t_ratio
3	-0.108	0.390	-0.276

Effect	Estimate	SE	t_ratio
4	1.641	0.205	7.992
5	0.350	0.038	9.158
6	-0.376	0.079	-4.760
7	-0.044	0.023	-1.908
8	0.266	0.112	2.388

Convergence was achieved in the final run ($t_{\text{conv.max}} = 0.122 < 0.25$ and $\max(|t_{\text{conv}}|) = 0.032 < 0.10$), so we proceed with interpretation of the estimated parameters

9. Model Fit

Goodness-of-fit (GoF) assesses whether the fitted SAOM can reproduce important structural features of the observed network. We simulate multiple networks from the fitted model and compare observed network statistics to the distribution of the same statistics in the simulated networks. If the observed statistics are not extreme relative to the simulated distribution (and the Monte Carlo p-values are not close to 0 or 1), the model is considered to fit those features reasonably well. We evaluate GoF for the indegree distribution, outdegree distribution, and the triad census for our final model from Question 8 (Model 1). We then update the model once (Model 2) by adding an additional endogenous effect (**reciAct**) and re-run GoF to assess whether fit improves.

Note: some statistics are not plotted because their variance is 0.
This holds for the statistics: 10 11 12 13 14 15.

Note: some statistics are not plotted because their variance is 0.
This holds for the statistics: 12 13 14 15.

Table 13: GoF Monte Carlo p-values for Model 1.

Statistic	P_value
Indegree distribution	NA
Outdegree distribution	NA
Triad census	NA

Figure 6: Goodness-of-fit for the indegree distribution (Model 1).

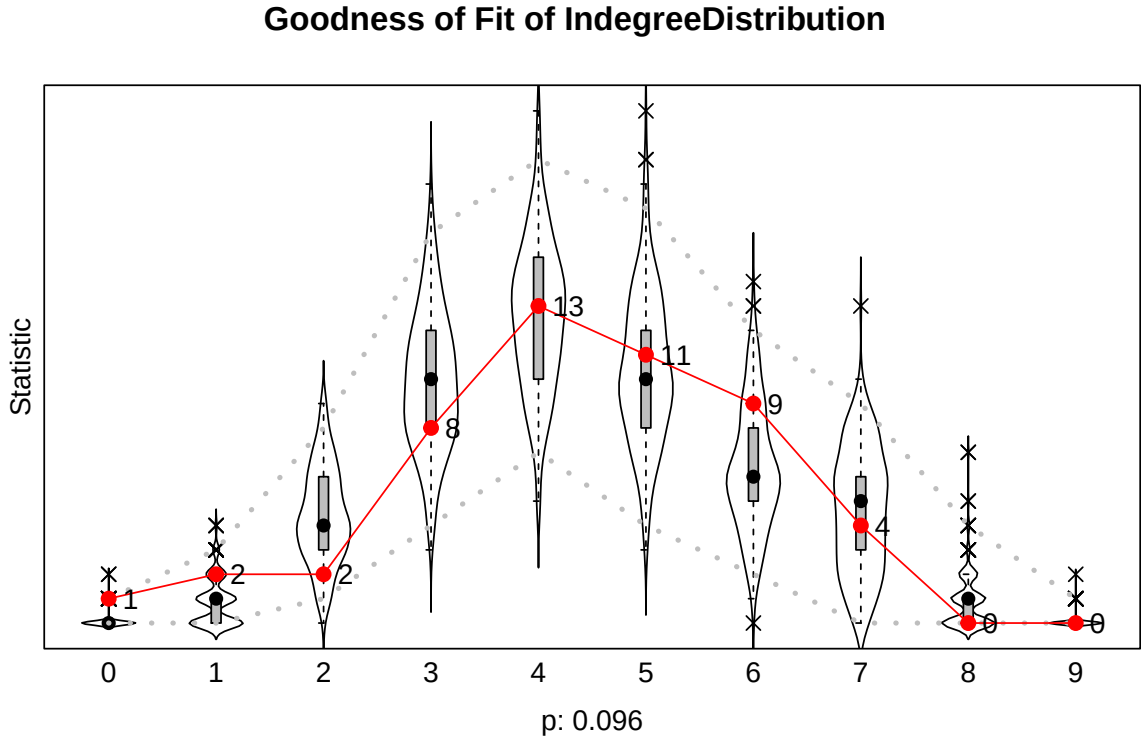


Figure 7: Goodness-of-fit for the outdegree distribution (Model 1).

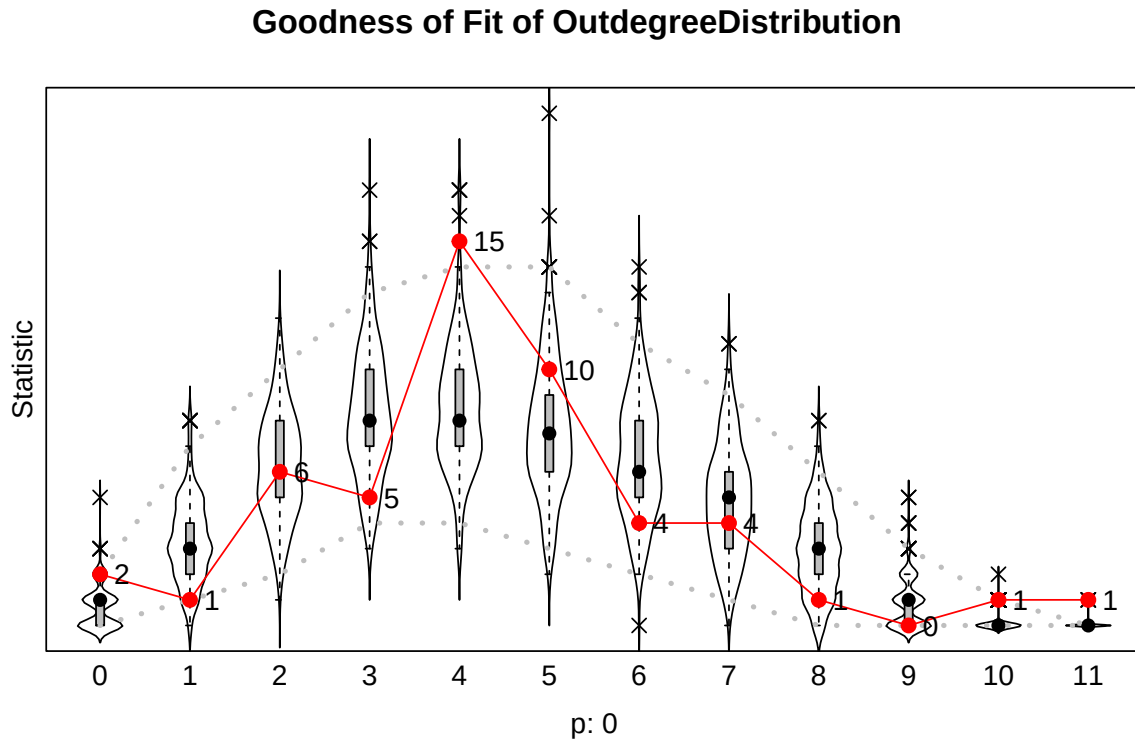
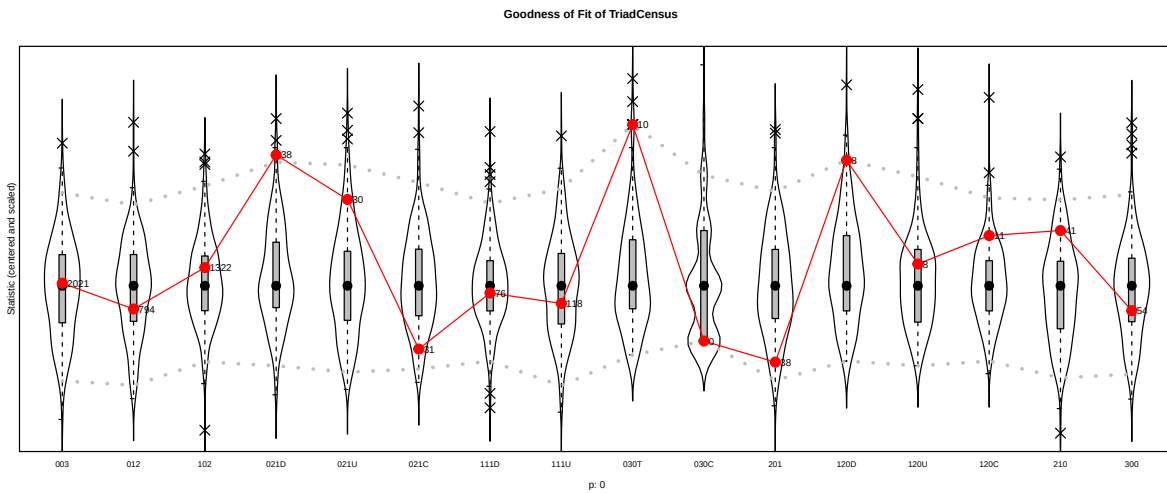


Figure 8: Goodness-of-fit for the triad census (Model 1).



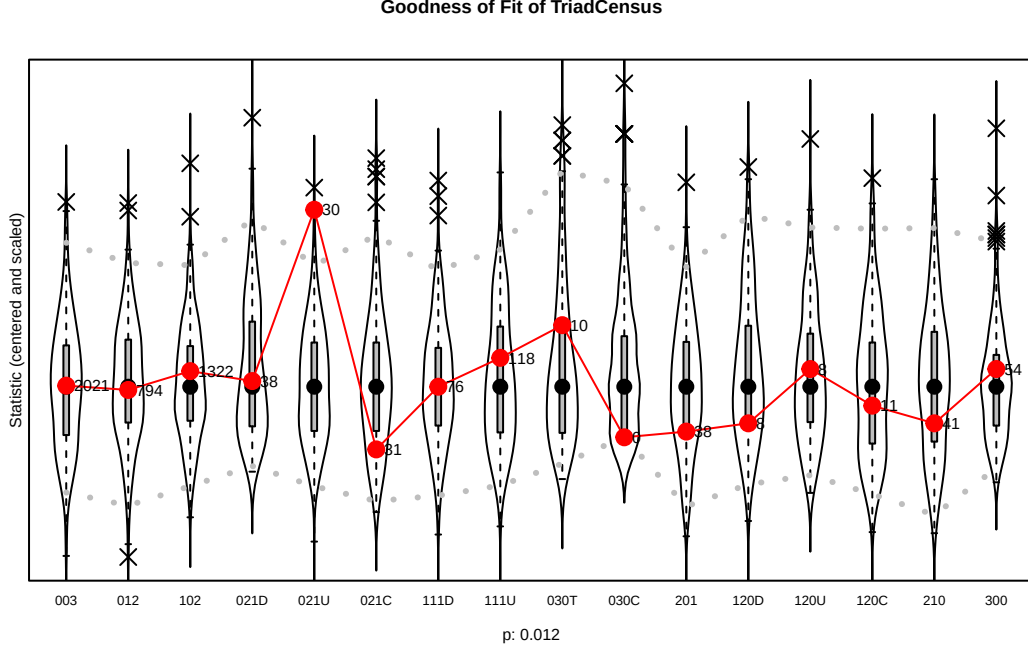


Table 14: Comparison of GoF Monte Carlo p-values (Model 1 vs. Model 2).

Statistic	P_Model1	P_Model2
Indegree distribution	NA	NA
Outdegree distribution	NA	NA
Triad census	NA	NA

10. Discussion and Limits

We interpret the converged SAOM estimates from Model 1 (three waves; constant node set). Convergence was achieved (overall maximum convergence ratio = 0.1221, with small parameter-wise convergence t-ratios), so the parameter estimates can be interpreted. In SAOMs, positive parameter estimates indicate that a mechanism increases the attractiveness of creating/maintaining a tie, whereas negative estimates indicate that a mechanism discourages tie formation/maintenance, conditional on the other effects in the model. **Baseline network tendencies.** The density (outdegree) parameter is negative (Estimate = -0.108, SE = 0.390), consistent with a sparse directed nomination network. Reciprocity is strongly positive (Estimate = 1.641, SE = 0.205; $t = 7.99$), indicating that best-friend nominations are frequently mutual. **H1 (Dyadic): Sex homophily.** The same-sex effect (sameX with sex_cov) is positive (Estimate = 0.266, SE = 0.112; $t = 2.39$). This supports the hypothesis that pupils are more likely to nominate

same-sex classmates as best friends, controlling for other network tendencies. **H2 (Triadic): Transitivity.** The transitive triplets effect is positive and highly statistically meaningful (Estimate = 0.350, SE = 0.038; $t = 9.16$). This provides strong evidence for triadic closure: best-friend ties tend to form in ways that close transitive structures (“friends of friends become friends”). **H3 (Node-level): Popularity reinforcement.** Contrary to our expectation, indegree popularity (inPop) is negative and statistically meaningful (Estimate = -0.376, SE = 0.079; $t = -4.76$). This does not support a Matthew-effect interpretation in this classroom; instead, conditional on reciprocity and transitivity, pupils appear less likely to nominate classmates who already receive many nominations. A plausible substantive interpretation is that “best friends” nominations reflect relatively stable, reciprocated friendship groups rather than accumulating nominations onto already popular pupils. In small classroom networks, high indegree may also be constrained by the nomination design (capped nominations), leaving less room for additional incoming nominations. **Additional control.** Outdegree activity (outAct) is slightly negative (Estimate = -0.044, SE = 0.023; $t = -1.91$), suggesting modest heterogeneity in sending nominations. Goodness-of-fit results for Model 1 indicate acceptable fit for the indegree distribution ($p = 0.096$), but clear misfit for the outdegree distribution ($p = 0$) and the triad census ($p = 0$). Following a common strategy for triad-related misfit, we updated the model (Model 2) by adding reciAct. The triad-census GoF improves ($p = 0.012$), indicating some improvement, although misfit concerns may remain for higher-order structure. Importantly, the substantive conclusions about our core hypotheses are stable across models: transitivity remains positive and strong, and inPop remains negative. In Model 2, reciAct is negative (Estimate = -0.460, SE = 0.104), suggesting the update captures additional reciprocated-degree structure that was not well reproduced by Model 1. This analysis has several limitations. First, to meet SAOM requirements we restricted the analysis to pupils present in all three waves, reducing the sample and potentially excluding relevant friendship dynamics involving movers or joiners. Second, best-friend nominations rely on self-reports and are constrained by the survey design (maximum nominations), which can affect degree-related mechanisms like popularity. Third, despite the model update, GoF indicates remaining misfit (especially for outdegree and triads in Model 1), suggesting that additional endogenous effects (e.g., alternative degree-dependence terms) or additional covariates could further improve fit and explanation. Finally, using only three waves limits temporal resolution; if classroom composition is sufficiently stable, adding an additional wave could increase power and help stabilize estimates.